Ingresso e protezioni

File: ingresso.kicad_sch

Canali di potenza

File: canali.kicad_sch

Controllo

File: controllo.kicad_sch

Alimentazione

File: alimentazione.kicad_sch

Carico elettronico 4 canali — 60 V / 20 A / 300 W

Ogni canale: 0–5 A, max 75 W per MOSFET. Sopra 15 V il limite e la potenza (300 W / V), gestito da firmware + ALERT INA228.

Setpoint: DAC8565 0–2,5 V –> partitore 25:1 –> 0–100 mV su shunt 20 mOhm = 0–5 A. Il partitore e riferito al Kelvin freddo dello shunt (SHNx).

ARM (attivo alto, pull–down 10k) abilita il carico. LM393 (OVP 64,7 V, OTP ~100 C) e ALERT degli INA228 lo tirano basso in hardware.

ARM basso –> CLAMP alto –> i FET di clamp cortocircuitano VSET1..4 a massa. MCU in reset = carico spento.

GND unica = DUT– (stella sul prigioniero J2). Controllo solo via WiFi; USB del Wemos NON isolato: usarlo solo per programmare a DUT scollegato.

Firmware ESP8266: dopo SPI.begin() riassegnare GPIO12 come uscita (ARM_DRV); GPIO16 = ARM_FB; A0 = NTC1 (partitore interno 220k/100k del D1 mini).

Pannello (rev 0.3): PCF8574 @0x20 (encoder A/B/SW, tasto OUTPUT, LED), INT su GPIO0; OLED @0x3C sullo stesso bus. OUTPUT ON abilita ARM, mai automatico.

Eccezioni alla regola SMD SOIC/SOP + passivi 1206

MOSFET di potenza TO–247 (montati sullo spreader), shunt WSK2512 Kelvin, diodo inversione D2PAK, TVS SMC/SMA,

INA228 in VSSOP–10 (passo 0,5 mm), DAC8565 in TSSOP–16, AP63205 in TSOT–23–6, induttore SRN4018, connettori THT.

Da verificare prima del layout

Antenna del Wemos fuori dal bordo e lontana dal cooler (o antenna esterna su U.FL), stabilita del loop CC verificata in analisi (C 1n C0G / R 1k / Rg 100R, rev 0.2), da confermare al banco, I2t di D1 contro F1, FBSOA IXTH75N10L2 a 60 V e Tc 90 C.

Schema gerarchico: un foglio per blocco funzionale

IV3HGA

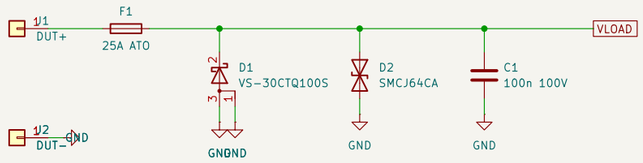
Sheet: /

File: eload_4ch.kicad_sch

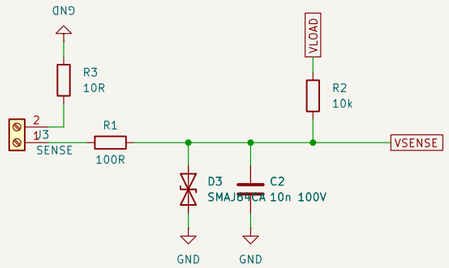
Title: Carico attivo 60V 20A 300W — Indice

Size: A3	Date: 2026–10–25	**Rev: 0.3**
KiCad E.D.A. 10.0.7		Id: 1/5

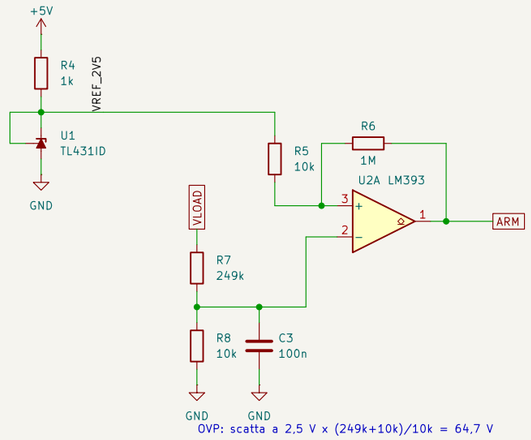
Morsetti DUT e protezioni di potenza



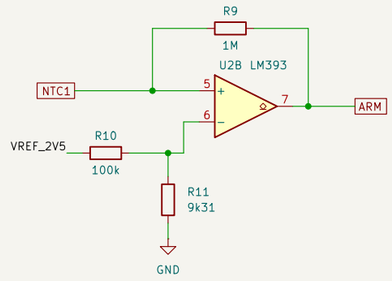
Remote sense (tensione misurata dagli INA228)



Riferimento 2,5 V e comparatori OVP / OTP (uscite open-collector su ARM)

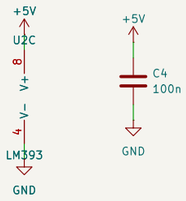


OVP: scatta a 2,5 V x (249k+10k)/10k = 64,7 V



OTP: NTC 10k B3950 a ~100 C = 0,21 V < OTP_REF

Alimentazione comparatore



Morsetti DUT, fusibile, diodo inversione, TVS, remote sense, OVP/OTP hardware

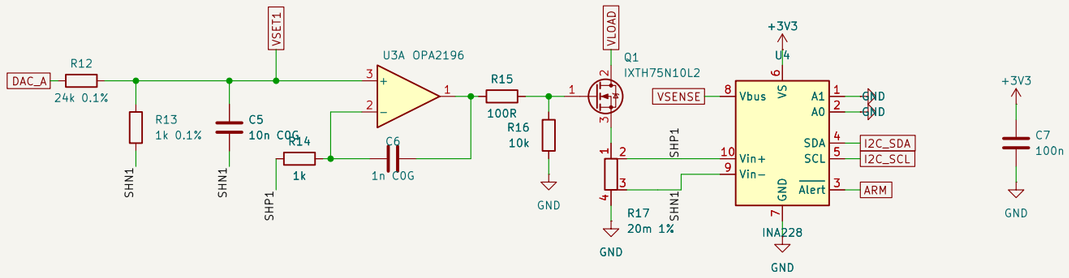
IV3HGA

Sheet: /Ingresso e protezioni/
File: ingresso.kicad_sch

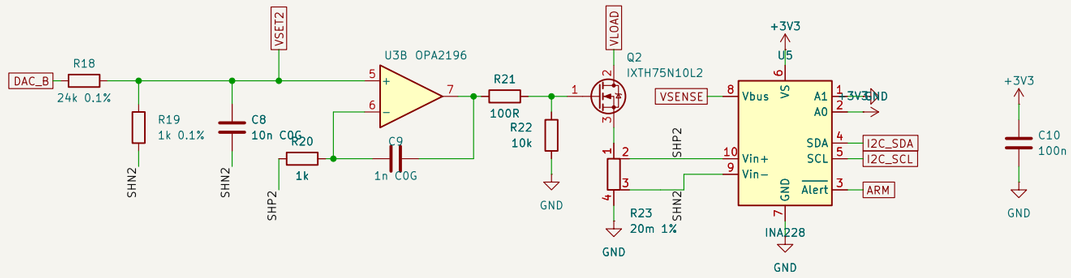
Title: Carico attivo 60V 20A 300W — Ingresso e protezioni

Size: A3	Date: 2026-10-25	**Rev: 0.3**
KiCad E.D.A. 10.0.7		Id: 2/5

Canale 1 — 0–5 A, 75 W — INA228 @0x40



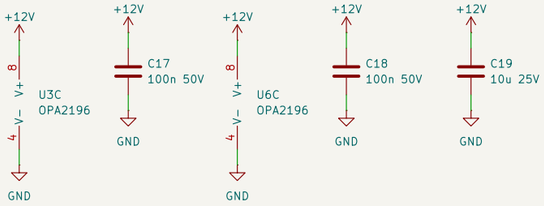
Canale 2 — 0–5 A, 75 W — INA228 @0x41



Canale 3 — 0–5 A, 75 W — INA228 @0x42

Canale 4 — 0–5 A, 75 W — INA228 @0x43

Alimentazione op-amp



4 canali CC: setpoint, loop OPA2196, MOSFET lineare, shunt Kelvin, INA228

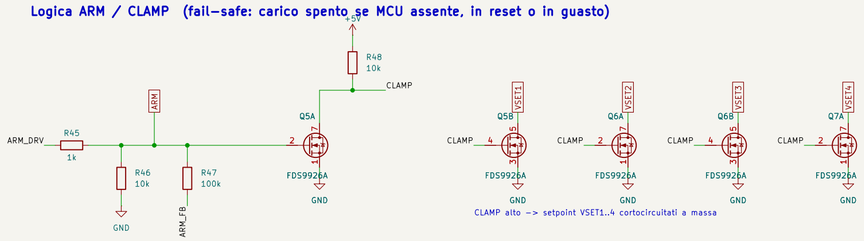
IV3HGA

Sheet: /Canali di potenza/
File: canali.kicad_sch

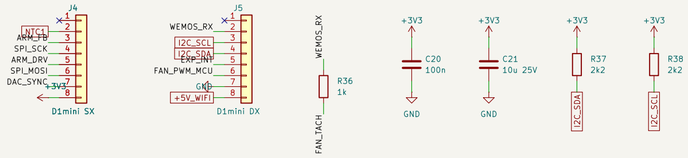
Title: Carico attivo 60V 20A 300W — Canali di potenza

Size: A3	Date: 2026-10-25	**Rev: 0.3**
KiCad E.D.A. 10.0.7		Id: 3/5

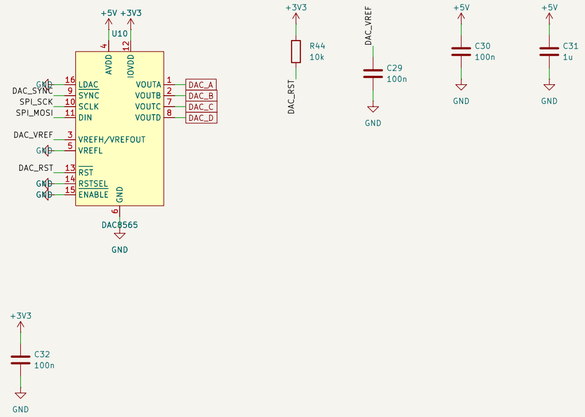
Logica ARM / CLAMP (fail-safe: carico spento se MCU assente, in reset o in guasto)



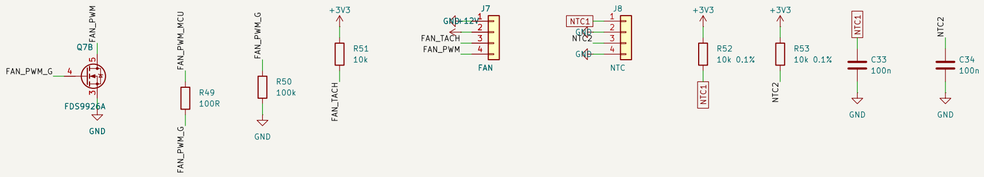
Modulo WiFi Wemos D1 mini Pro (ESP8266) su strip femmina



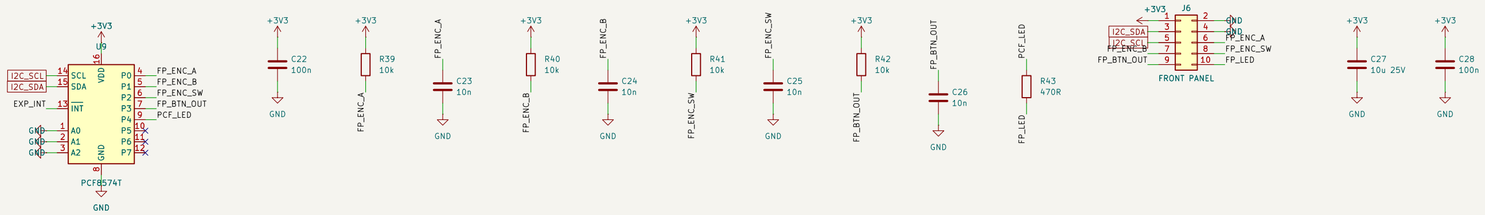
DAC quadruplo 16 bit (riferimento interno 2,5 V)



Ventola 4 fili PWM e sensori NTC sullo spreader



Pannello frontale: OLED I2C, encoder con pulsante, tasto OUTPUT (rev 0.3)



Wemos D1 mini Pro (ESP8266), DAC8565, logica ARM/CLAMP, ventola, NTC

IV3HGA

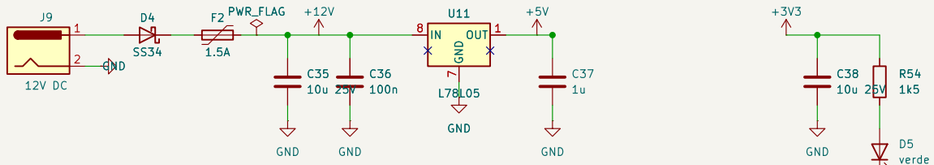
Sheet: /Controllo/
File: controllo.kicad_sch

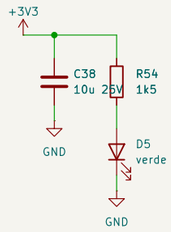
Title: Carico attivo 60V 20A 300W — Controllo

Size: A2	Date: 2026-10-25	**Rev: 0.3**
KiCad E.D.A. 10.0.7		Id: 4/5

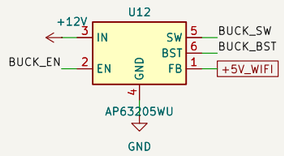
Ingresso ausiliario 12 V (alimentatore isolato dedicato) e regolatori

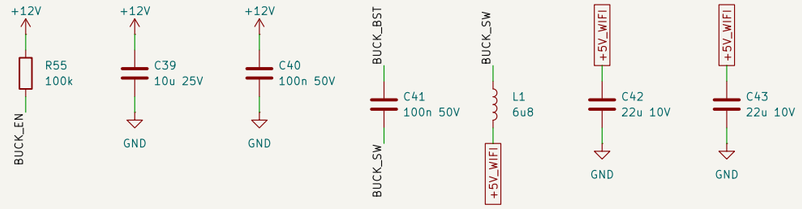
+3V3 fornita dal regolatore del modulo Wemos (pin 3V3)

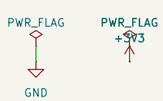




Buck 12 V –> 5 V (AP63205) solo per il Wemos: il 5 V analogico resta sul 78L05







12 V ausiliaria, 5 V analogico (78L05), buck 5 V per il WiFi

IV3HGA

Sheet: /Alimentazione/
File: alimentazione.kicad_sch

Title: Carico attivo 60V 20A 300W – Alimentazione

Size: A3	Date: 2026-10-25	**Rev: 0.3**
KiCad E.D.A. 10.0.7		Id: 5/5